

Requirements for testing

1. DSO
2. Windows-based CRO software
3. USB sound card
4. Audio cable for connection with PC
5. AUX cable for connection with DSO

Function Generator app installed on Android smartphone along with audio and crocodile cables for connection to DSO is shown in Fig. 1.



Fig. 8: DSO used to observe output of the signal generator from Android app

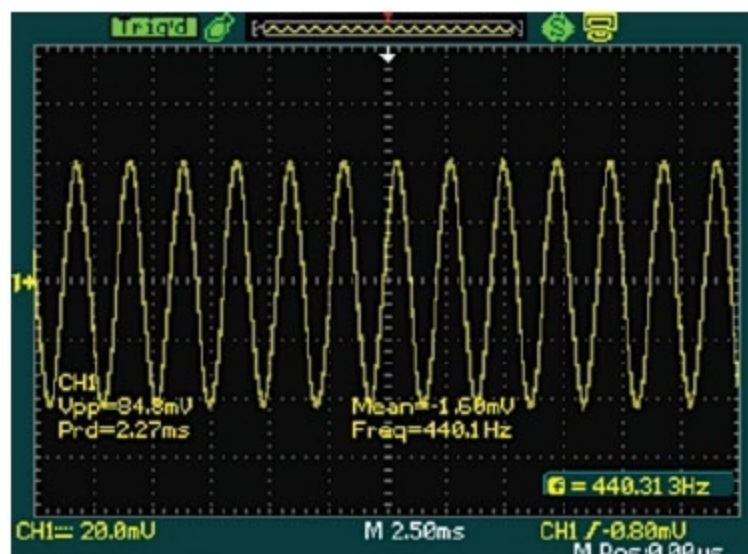


Fig. 9: 440Hz sine wave output as observed on oscilloscope

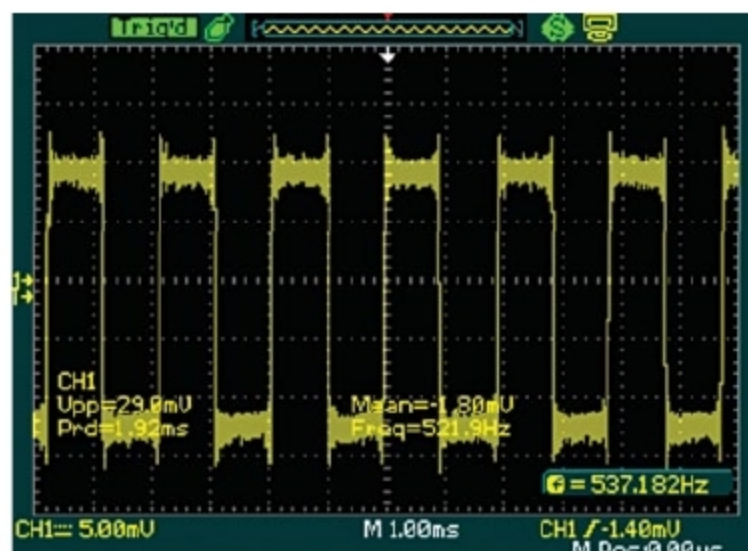


Fig. 10: 537Hz square-wave output as observed on oscilloscope

Steps to perform signal generation

Function Generator app. Download Function Generator app by Keuwlsoft from Play Store, and install it on your smartphone. Function Generator app running on cellphone is shown in Fig. 2.

1. Select channel for signal (CH1 or CH2).
2. Select waveform, such as sine, triangular or square. Select CH1 (channel) and sinewave (waveform), as shown in Fig. 3.

3. Select FREQ block and adjust frequency using slider or text input box.

4. Select OUT from OUTPUT section, as shown in Fig. 4.

5. Corresponding function output signal will be available at audio OUT of the cellphone. You will be able to hear the frequency sound in the loudspeaker of your phone or headphone if connected to the phone.

6. You can use a 3.5mm audio jack at cellphone audio OUT to connect to an external device, as mentioned above.

7. You can add noise to the generated signal.

8. Options are also available for modulating the waveform using FM or AM, as shown in Fig. 5.

9. VOL is used for increasing or decreasing the volume in OUT section.

10. You can provide range in output signal using START and STOP options in SWEEP

section. The slider can be used to change mode in SWEEP settings.

AM signal of amplitude-modulated sinewave on DSO is shown in Fig. 6.

Sound Generator app. Sound Generator app (from Alok Mandavgane) is another option for sawtooth-wave, square-wave, triangular-wave and sinewave signals. White noise generation option is also available on this app. Output is available at the phone's loudspeaker.

The app has a simple interface and is easy to understand. Frequency and volume control sliders are provided at the top of the app, and function type settings are provided at the bottom, as shown in Fig. 7.

It supports signal range from 20Hz to 20,000Hz. Select frequency and switch on/off button to activate the signal and then select the type of wave like sinewave, square-wave and so on. A 522Hz square-wave signal generated using Sound Generator app is shown in Fig. 7.

Testing the signals on DSO

Signals can be tested or analysed on a standalone digital oscilloscope, or PC-based oscilloscope software. Follow the steps given below for testing the signals on DSO. Connections with DSO are shown in Fig. 8.

1. Connect DSO with Android phone's 3.5mm jack using suitable cables (aux and crocodile).

2. Check values and signal waveforms on DSO.

3. 440Hz sinewave and 537Hz square-wave signals generated by Signal Generator app as observed on DSO are shown in Fig. 9 and Fig. 10, respectively. (Note that, 3.5mm jack aux cable can cause some distortions in signals on DSO.)

Testing the signals on PC

You can also use PC-based CRO software, such as Zelscope Demo Version, to analyse the generated signals mentioned above. Frequency signal values and waveforms are obtained on Zelscope software.

A USB sound card is used in this case, as it minimises any damage caused to the internal sound card of the PC if a high-value signal is applied. Options to save waveform images and

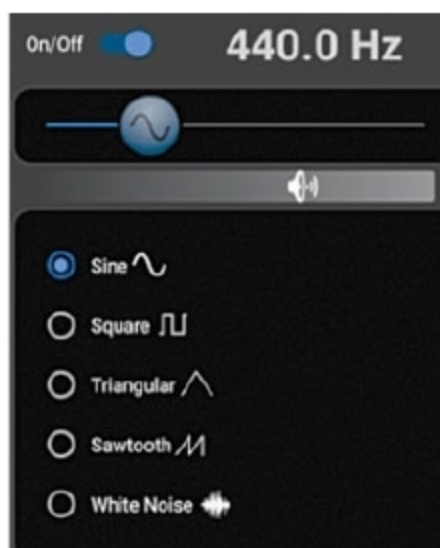


Fig. 11: 440Hz sinewave signal